

TREKKING MANAGEMENT SYSTEM REPORT

1. Student Details

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About Me: I am a Diploma level student in the IIT Madras BS Degree Program with a keen interest in Web Development, Artificial Intelligence, Machine Learning, and Generative AI. I enjoy developing practical applications that integrate modern technologies to solve real-world problems while continuously improving my software development skills.

2. Project Details

Project Title: Trekking Management System

Problem Statement: To design and build a web-based Trekking Management System that enables administrators, trek staff, and trekkers to **manage** trekking activities efficiently. The application allows users to **search** and **book** treks, while administrators and staff manage treks, bookings, and participants. It also prevents overbooking, **tracks** trek status, and maintains complete trekking history.

Approach: The application was developed using Flask with a modular blueprint architecture and SQLite for database management. It provides separate dashboards for Admin, Staff, and Users, enabling trek management, booking, participant management, staff assignment, and secure role-based authentication through a responsive web interface.

3. AI/LLM Declaration

I used **ChatGPT** to assist in writing SQLAlchemy model definitions, creating API documentation samples, and improving variable naming consistency.

The extent of AI/LLM usage is around **15–18%**, limited to **code suggestions and documentation formatting**.

All final implementation logic, debugging, and integration were done manually.

4. Technologies and Frameworks Used

Technology /Library Purpose

Flask	Backend web framework
SQLAlchemy	Database ORM
SQLite	Local database
Jinja2	HTML template engine
Bootstrap 5	Responsive UI design
Flask-Login	User authentication and session management
Werkzeug Security	Password hashing
HTML5 & CSS3	Frontend structure and styling

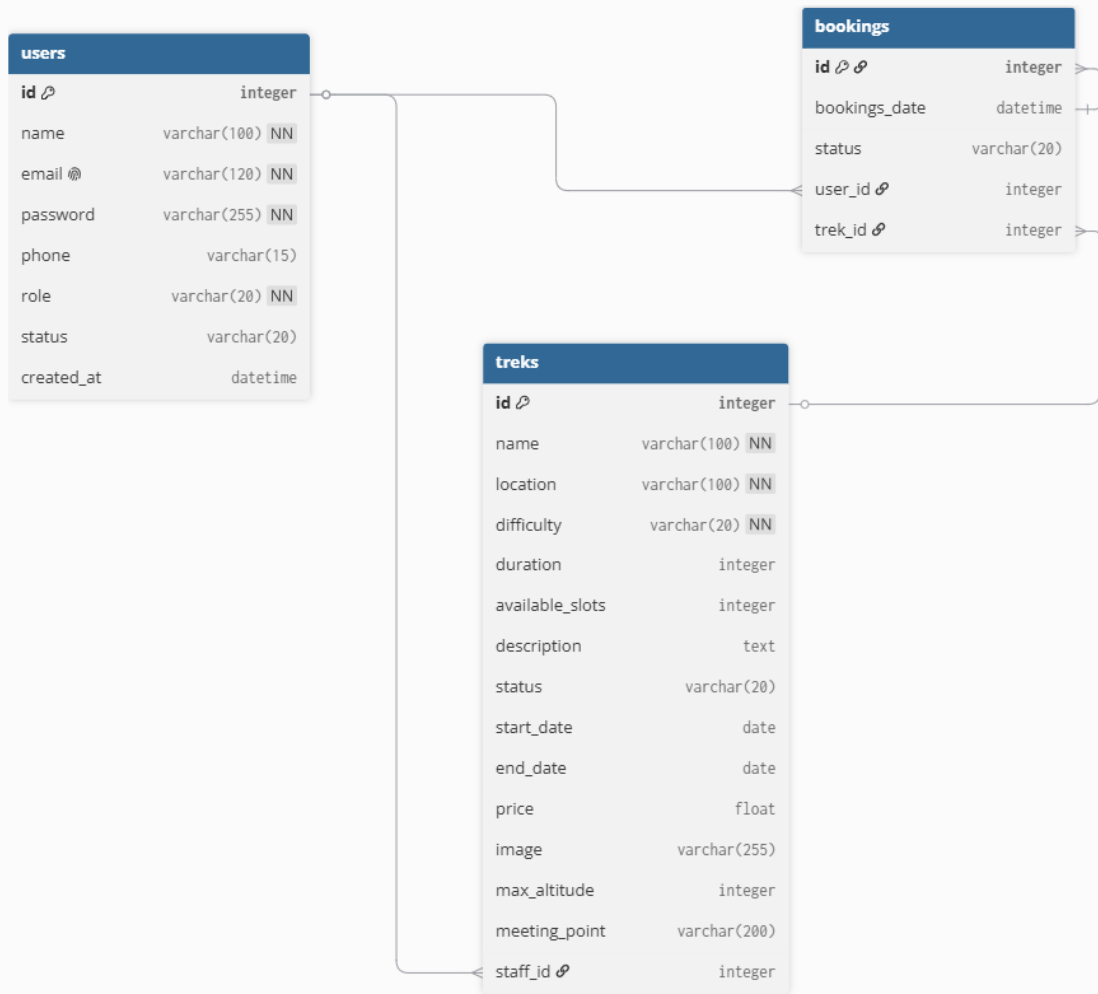
5.Database Schema / ER Diagram

Tables:

1. **User**— Stores user information (*id, name, email, password, phone, role, status, created_at*).
2. **Trek** — Stores trek details (*id, name, location, difficulty, duration, available_slots, description, status, start_date, end_date, price, image, max_altitude, meeting_point, staff_id*).
3. **Booking** — Stores trek booking records (*id, booking_date, status, user_id, trek_id*).

Relationships:

- **One-to-Many** → **User** → **Booking** (One user can have multiple bookings.)
- **One-to-Many** → **Trek** → **Booking** (One trek can have multiple bookings.)
- **One-to-Many** → **Staff (User)** → **Trek** (One staff member can manage multiple assigned treks.)



6. API Resource Endpoints

Endpoint	Method	Description
/auth/login	POST	Authenticate user and create a login session
/user/treks	GET	Retrieve all available treks
/user/book/<trek_id>	GET	Book a selected trek
/admin/treks/add	POST	Add a new trek (Admin only)

YAML API Definition File:

Included separately in the submission ZIP as api.yaml.

7. Architecture and Features

Architecture Overview

- **app.py** – Main Flask application entry point.
- **app/models.py** – SQLAlchemy database models.
- **app/auth, app/admin, app/user, app/staff** – Flask Blueprints for role-based modules.
- **app/templates** – Jinja2 templates for dynamic web pages.
- **app/static** – CSS, images, and static assets.

Implemented Features

- User registration and secure login.
- Role-based access control (Admin, Staff, User).
- Trek management (Add, Edit, Delete).
- Trek booking and cancellation.
- Staff assignment and participant management.
- Trek search and filtering by location and difficulty.
- User profile and password update.
- Responsive user interface using Bootstrap 5.

Additional Features

- Booking history for users.
- Dashboard for Admin, Staff, and Users with role-specific information.
- Secure password hashing and session management using Flask-Login and Werkzeug Security.

8. Video Presentation

Drive Link: <https://drive.google.com/file/d/12RAQiN1cOR5fCOplI9Zo3-IFVUpAEaW5/view?usp=sharing>

Github link: <https://github.com/MRIGAANKSh/iitm-mad1-project.git>

